Soil microbes create humus: Humus is the dark organic matter in soil. The humus is formed when dead plant and animal matter are decayed by soil microorganisms. Humus has many nutrients that improve the fertility of soil, Nitrogen being the most important.

Humus helps soil to retain moisture, and encourages the formation of soil structure. Soil organisms promote plant growth.

Soil Algae

Soil algae are ubiquitous in nature wherever moisture and sunlight are available. They are visible to the unaided eye in the form of green scum on the surface of soils. Numerically, they are not as many as Fungi, Bacteria or Actinomycetes. Some of the common algae in Indian soil are *Chlorella, Chalmydomonas, Chlorochytrium, Chlorococcum* and *Oedogonium*.

Blue green algae, or *Cyanophyceae,* are responsible for Nitrogen fixation. The amount of Nitrogen they fix depends more on physiological and environmental factors rather than the organism's abilities. These factors include intensity of sunlight, concentration of inorganic and organic Nitrogen sources and ambient temperature and stability.

Soil Protozoa

Soil protozoa are unicellular. They are characterized by a cyst in their life cycle which can help the species to withstand adverse soil conditions. The protozoans prefer certain species of bacteria for their nutrition. Protozoa are abundant in the upper layer of the soil and their numbers are directly dependent on bacterial population.

HOTS

"Without fungi even death will be incomplete" - Pasteur

Justify the statement.

10.4.1 Factors Influencing Microbial Population in Soil

The major factors that influence the microbial community in soil are

- Moisture
- pH
- Temperature
- Gases
- Organic and inorganic fertilizer
- Organic matter of soil
- Types of vegetation and growth stages
- Ploughing
- Season
- Depth of soil

Infobits

Some soil microbes produce a variety of substances that promote plant growth, including Auxins, Gibberellins and antibiotics.

10.5 Microbial Interactions

Microorganisms in soil interact with themselves and lead to beneficial and

TN_GOVT_XI_Micro_Biology_CH10.indd 153 16-02-2019 10.03.51 AM